

# JEE MAIN REVISION MASTER: SALT ANALYSIS

## 2026 1 QUESTIONS 1.67% WEIGHTAGE ↓49.85%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

### 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                               | Frequency | Category              |
|---------------|------------------------------------------------------------|-----------|-----------------------|
| Standard      | Salt Analysis — Standard Calculations & Problem Archetypes | 2 PYQs    | Quantitative Analysis |
| Standard      | Salt Analysis — Core Principles & Governing Laws           | 1 PYQs    | Core Principle        |
| Standard      | Salt Analysis — Applied Cases, Reactions & Traps           | 1 PYQs    | Applied Topics        |

### 2. Master Formula Reference Card

| Concept / Topic                                            | Mathematical Formulation & Governing Equations                                     |
|------------------------------------------------------------|------------------------------------------------------------------------------------|
| Salt Analysis — Standard Calculations & Problem Archetypes | $\text{NH}_4\text{Cl(s)}, \quad \text{NH}_4\text{OH}, \quad \text{H}_2\text{SO}_4$ |
| Salt Analysis — Applied Cases, Reactions & Traps           | $\text{NH}_3, \quad \text{NH}_4\text{Cl(s)}$                                       |

### 3. Deep-Dive Concepts & Linked PYQ Evidence

#### 3.1 Salt Analysis — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

**Core Principle:** Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Salt Analysis.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "When salt BA is treated with Conc.  $\text{H}_2\text{SO}_4$  reddish brown gas is liberated. The aqueous solution of BA gives pale yellow ..." [Ans: D]
- [2024 Shift] Q#4 (MCQ): "In the analysis of III group basic radicals of salts, the purpose of adding  $\text{NH}_4\text{Cl(s)}$  to  $\text{NH}_4\text{OH}$  is..." [Ans: N/A]

### 3.2 Salt Analysis — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

**Core Principle:** Fundamental definitions, core postulates, and governing physical/chemical relations in Salt Analysis.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Salt Analysis.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Salt Analysis PYQs.

**Representative Source PYQs:**

- **[2025 Shift] Q#3 (MCQ):** "The sodium fusion extract is boiled with concentrated nitric acid while testing for halogens. By doing so, it..." [Ans: N/A]

### 3.3 Salt Analysis — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

**Core Principle:** Special cases, experimental observations, and conceptual traps designed to test deep understanding of Salt Analysis.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

**Representative Source PYQs:**

- **[2025 Shift] Q#2 (MCQ):** "The group reagent  $\text{NH}_4\text{Cl(s)}$  and aqueous  $\text{NH}_3$  will precipitate which of the following ion?..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

When salt BA is treated with Conc.  $\text{H}_2\text{SO}_4$  reddish brown gas is liberated. The aqueous solution of BA gives pale yellow precipitate with  $\text{AgNO}_3$  solution. Which of the following anion ( $\text{A}^-$ ) is present in the salt BA?

- (A)  $\text{Cl}^-$
- (B)  $\text{CO}_3^{2-}$
- (C)  $\text{SO}_4^{2-}$
- (D)  $\text{Br}^-$  [Correct]

**Correct Answer: D**

**Solution Analysis:** The anion present is  $\text{Br}^-$ . When a bromide salt is treated with concentrated  $\text{H}_2\text{SO}_4$ , reddish brown vapours of bromine are liberated. First, hydrogen bromide is formed:  $\text{BBr} + \text{H}_2\text{SO}_4 \rightarrow \text{BHSO}_4 + \text{HBr}$ . Then HBr reduces concentrated  $\text{H}_2\text{SO}_4$  and gets oxidised to bromine:  $2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ . The bromine vapours,  $\text{Br}_2$ , are reddish brown in colour. Also, the aqueous solution of the salt gives a pale yellow precipitate with  $\text{AgNO}_3$  solution.